OXFORD

Interacting with Computers, 2023, **00**, 1–10
https://doi.org/10.1093/iwc/iwad029
Special issue paper

Global Virtual Team Working during the Covid-19 Pandemic

Phelim Barry[1] and Bridget Kane[2,*]

[1]Business School, Dublin City University, Dublin 9, Ireland
[2]Information Systems, Karlstad University Business School, Universitetsgatan 2, SE-651 88 Karlstad, Sweden
*Corresponding author: Bridget.Kane@kau.se

Abstract

Virtual work introduces distinct challenges when compared to face-to-face or on-site work. Communication and collaboration are key factors in team development and in team performance. When teamwork is of a global nature, imposed because of a pandemic, then additional variables are introduced into the equation. The challenges that global virtual teams (GVTs) encounter when communicating and developing in the context of the Covid-19 pandemic impacts how work is structured and teams develop. This qualitative study was conducted just over 1 year after the start of the Covid-19 pandemic when working from home became mandatory and particular challenges for GVTs became apparent. Data are gathered through an online anonymous survey and followed by semi-structured video-mediated interviews with staff in a large multinational software development company. Findings show that GVTs encounter a number of distinct challenges than face-to-face teams. However, individuals are actively adapting to the situation in which they find themselves and are learning to deal with the challenges by being proactive. Even though the tools used previously may not be entirely suitable for virtual work, GVTs learn to change how tools are used to be more productive and collaborative. Challenges continue to exist in areas, such as knowledge gathering and knowledge sharing. Communication failures can lead to delays and confusion. The findings also indicate that even though GVTs are not communicating as much informally or socially with their colleagues because of remote working in the pandemic, being remote is not having a negative effect on the ability to get work done. Some aspects of team development can be negatively affected when working in GVTs. Impacts are identified relating to trust and relationship building, as well as being able to identify and resolve conflict. These results prompt further research opportunities as organizations and individuals continue to adapt and embrace global virtual teamwork.

RESEARCH HIGHLIGHTS:

- A qualitative study of Global Virtual Teams (GVTs) and their challenges during the Covid-19 Pandemic.
- GVT members became proactive in dealing with the challenges of 100% remote working.
- GVTs adapt their methods of working and the tools available to them.
- Communication failures, building trust and relationships, and resolving conflict are significant challenges.
- Lack of informal communication opportunities adversely affects relationships in GVTs.
- Further investigation of gender differences in attitudes and practices are prompted; and the associations between trust and conflict resolution in virtual relationships.

Keywords: Covid-19 pandemic, global virtual teams, communication, teamwork

1. INTRODUCTION

The Covid-19 pandemic provides the opportunity to gather and analyse real world data to present a point in time view of global virtual team (GVT) operations. This research investigates the challenges that GVTs encounter when communicating and developing in the context of the pandemic. It also investigates the impact on how work is structured and on how teams develop. This qualitative study avails of the opportunity afforded by the Covid-19 pandemic to gain insight into the functioning of GVTs.

We undertook this research to help understand where companies see challenges in developing GVTs and the difficulties experienced while working on virtual projects. Our aim is to identify opportunities to improve the experience of working as part of a GVT. Our results will assist companies in preparing for the future of work. Designers of technology will be prompted to develop communication tools so that GVT performance can be improved. This research was conducted in Spring 2021, just over 1 year after the Covid-19 pandemic began. At that time there was little newly published research into the communications challenges affecting GVTs, in particular when the choice to work remotely was not voluntary. We aim to obtain a deep understanding of the communications challenges affecting GVTs and the impact that working in a virtual environment has on team development and cohesion. The following two research questions are addressed: What team development issues are GVTs encountering during the Covid-19 pandemic? What are the communication challenges impacting GVTs during the Covid-19 pandemic?

GVTs present special challenges in terms of communications. Informal water-cooler conversations no longer take place. Creating relationships takes longer and generating trust is difficult

Received: November 10, 2021. **Revised:** November 26, 2022
© The Author(s) 2023. Published by Oxford University Press on behalf of The British Computer Society.
This is an Open Access article distributed under the terms of the Creative Commons Attribution License (http://creativecommons.org/licenses/by/4.0/), which permits unrestricted reuse, distribution, and reproduction in any medium, provided the original work is properly cited.

Downloaded from https://academic.oup.com/iwc/advance-article/doi/10.1093/iwc/iwad029/7136028 by Karlstads universitet user on 12 May 2023

as communications generally tend to be more formal (Morrison-Smith and Ruiz, 2020b). Organizations need to align their technology and systems to support GVTs to enable them to share information effectively and develop relationships and trust (Suchan and Hayzak, 2001).

As we are awash with a multitude of new tools and applications to help us communicate with each other, it is not obvious if these applications are enhancing our communication processes or just adding additional layers to it. It is argued that some of these less rich, i.e. nonverbal, forms of communication can sometimes reduce ambiguity and enhance communication rather than negatively affect it (Scott and Fullwood, 2020).

The first research question investigates areas of team development, such as trust and co-operation, which can vary amongst GVTs but tend to increase if team members have worked together previously (Morgan *et al.*, 2014), and it is suggested that there are a range of communication methods that may impact on how the team can develop and operate effectively.

As project management (PM), teams and work, in general, move towards a more global environment, the second research question is designed to identify how GVTs communicate and how the various methods of communication can affect the development of the team. Tuckman and Jensen 's (1977) traditional team development stages of forming, storming, norming and performing are still widely used today but present many challenges to the development of GVTs given that all team interaction is virtual using electronic means. Virtual and traditional teams share the common characteristic of good communications (Dakory and Abdou, 2009) but the asynchronous nature of GVTs brings additional challenges to the development of the team (Peters and Manz, 2007).

1.1. Background

This research was undertaken within a large multi-national software company with over 30,000 employees working in locations that span the globe. We focus on the PM organization which consists of product managers, business analysts, program and project managers, as well as other subject matter experts and administrative staff from multiple global locations.

Pre-Covid, activities, such as project kickoffs or workshops that required large participation, would typically be held centrally in one of the global locations with all participants traveling to that location and working collocated for a period of time. However, travel restrictions during the Covid-19 pandemic prevented such gatherings from happening and staff were instructed to work from home.

Working from home due to the Covid-19 pandemic brought a serious change to the way of working for the majority of people but was even more significant given the collaborative nature of PM work, in the multinational software development company.

2. METHOD

Our research is split into two separate stages. Firstly, all members of the PM organization were invited to participate in an anonymous survey, (see Appendix A for survey questions), which was designed to capture initial thoughts on the challenges posed in virtual work. The PM organization consisted of approximately 60 employees of which 28 completed the survey. Following piloting with several GVT members (five), the survey was available for a period of four weeks and was split into two distinct sections: Section 1 focused on Project and Work related Communications and Section 2 focused on Team Work and Team Communications.

The results of the survey were analysed and those results were used to generate interview questions for the second stage of our research. This second stage consists of semi-structured video-mediated interviews to facilitate in-depth discussions into the survey results and provide for more open ended conversations with the participants. Each interview took approximately 30 minutes and the recordings were pseudonymized and analysed. The interviews were broken into two sections and generally followed the same structure as the surveys, with the first half of the interview expanding on communications with GVT and the second half focused on team work and associated challenges.

A timeframe of 1 week was scheduled for the completion of the interviews directly following the close of the survey in order to preserve the validity of the survey data. During this time a total of five interviews were completed, at which point, it was decided to end the interview stage as no new opinions were being heard and we had reached data saturation.

Thematic analysis of the qualitative data generated from the interviews is used to identify themes and patterns in the responses. Appropriate statistical methods are used in analysis. Spearman's rho, ρ, also called Spearman's Rank Correlation coefficient, is the non-parametric measure used to measure associations between the variables measured on Likert (ordinal) scales. Pearson's Chi-square test of independence is used when testing categoric variables.

This study was ethically approved by the Dublin City University Business School Ethics committee and protected the anonymity of the participants throughout.

The original data can be made available on request until December 2024, two years after the study was conducted, in accordance with DCU policy.

3. RESULTS

3.1. Survey results

The survey was completed by 28 (47% of the target population) of whom 16 are male (57%) and 11 female (39%); one respondent preferred not to say. All except three of the respondents are older than 25 years (25, 90%). Twelve respondents (43%) report that over two-thirds of their workload is spent on team activities while another 11 (39%) report that team activities absorb between one-third and two-thirds of their time.

3.1.1. *Communications and methods*

The importance of communications to project success is either extremely or very important to all (100%) respondents with almost half (12, 42%) saying that communication is more difficult in a virtual environment. Knowing that a message has been received and understood is also either extremely or very important for all respondents. No statistically significant differences in the view of the importance of communication, nor in relation to gender, age category nor the amount of time spent in virtual team working, are identified. Email remains the most widely used method for communication; however, video calling is deemed to be the most effective communication method. Although no statistically significant differences are identified between the communication preferences for males and females, females report using more voice calls, than males (Chi Sq $p \leq 0.05$; Spearman's correlation ρ, $p \leq 0.05$). Results on the preferred mode of communication, as well as the preferred and most used modes, are provided in Table 1.

Knowledge gathering is more difficult than knowledge sharing in a GVT environment. Half of our respondents (14, 50%) find

Downloaded from https://academic.oup.com/iwc/advance-article/doi/10.1093/iwc/iwad029/7136028 by Karlstads universitet user on 12 May 2023

Table 1. Modes of communication used amongst GVTs during the Covid-19 pandemic

Communication Mode	Most Effective	Preferred	Most Used
	Frequency (%)		
Email	5 (18)	7 (25)	12 (43)
Text/IM	2 (7)	3 (11)	7 (25)
Voice Call	6 (21)	6 (21)	5 (18)
Video Call	10 (36)	6 (21)	3 (11)
Face-to-Face	5 (18)	6 (21)	1 (4)
Total	28 (100)	28 (100)	28 (100)

knowledge gathering is somewhat difficult in a GVT, while 10 (36%) find that knowledge sharing is somewhat difficult. No significant differences in views because of gender, age nor the amount of time spent in virtual team work are evident.

When asked if virtual projects can be run as effectively as face-to-face projects, almost half (13, 46%) say yes, that virtual can be as effective as face-to-face projects, a further 7 (25%) replied maybe and almost one-third (7, 29%) say no. When these results are examined for age category or gender differences no statistically significant differences are evident. However, those who spend more than two-thirds of their time in virtual team-working report that projects that are run remotely/virtually are not as effective as face-to-face projects (Chi Sq, $p \leq 0.05$; Spearman's correlation ρ, $p \leq 0.05$).

3.1.2. *Relationships and teamwork*

Building trust is considered more important than building relationships in a GVT and building strong relationships is believed to be difficult. Building relationships in a GVT is considered either very or extremely important by 25 (89%), while almost all say that building trust is either extremely or very important (27, 97%). Almost two-thirds (17, 61%) say that building strong relationships is somewhat or extremely difficult while over one-third (11, 39%) say that trust is somewhat or extremely difficult to build.

Table 2 shows the views of the importance of building trust and building relationships and the difficulties experienced for GVTs. It is considered easier to build trust in GVTs than it is to build relationships.

Although the importance of trust tends to be reported as extremely important more often for those who spend more than two-thirds of their time in team activity (see Table 3), the differences are not statistically significant. There is no statistically significant differences regarding the importance of trust, based on age category or gender. A future survey with a larger sample size may likely yield significant differences.

The three most important characteristics of GVTs are open communications (15, 54%); active participation (14, 50%) and trust (14, 50%), detailed in Figure 1, which also lists the three least important characteristics as diversity, accountability and having common goals.

Informal Conversations: Informal conversation plays an important role in building relationships. Informal conversations are either very or extremely important to relationship building (23, 82%), and since the start of the Covid-19 pandemic, there are much fewer informal interactions. Fewer informal conversations are reported by almost half of the respondents (13, 46%) than pre-pandemic, while a further 11 respondents (39%) report somewhat less. A strong association is demonstrated between the importance of informal conversations and the importance of relationship building (ρ, $p \leq 0.01$, one-tailed). The amount of informal conversation is not influenced by age group, gender or the amount of time spent in virtual team activities.

The three most serious challenges affecting GVTs are time zones (19, 68%), lack of personal connection (19, 68%) and communication failures (17, 61%), given in Table 4.

The three most difficult global virtual team (GVT) activities are i) building a cohesive team, (13, 48%), ii) resolving conflict (12, 43%) and iii) creating a healthy culture (12, 43%) as shown in Table 5. Overall, reaching agreements in GVTs is not considered overly difficult, however, significant differences are noted for those who spend more than two-thirds of their time in GVT activities and for those older than 25 years. Those who spend more than two-thirds of their time in GVT activity report more difficulty than those who spend less of their time in team activities (Chi Sq, $p \leq 0.05$). Those older than 25 years report reaching agreements somewhat more difficult than those under 25 (Chi Sq $p \leq 0.05$). There is no statistically significant difference depending on gender.

There are mixed views on the ease or difficulty experienced in being creative, and none of the differences are dependent on gender, age category or the amount of time spent in GVT activities. Indeed, reaching agreements (10, 36%) and being creative (9, 32%) are deemed to be the easiest activities to complete virtually.

Team Cohesion: Difficulties with building cohesive teams are linked to several variables, namely trust, building relationships, knowledge gathering and knowledge sharing. Difficulties in building a cohesive team is associated with the difficulty of building trust (ρ, $p \leq 0.01$). There is also a significant association

Table 2. Trust and Relationships in Global Virtual Team (GVT) Development

	Not Important	Slightly Important	Moderately Important	Very Important	Extremely Important
			Frequency (%)		
Importance of Building Trust	0	1 (3)	0	13 (46)	14 (50)
Importance of Building Relationships	0	0	3 (11)	9 (32)	16 (57)
	Extremely Difficult	**Difficult**	**Neither Easy nor Difficult**	**Somewhat Easy**	**Extremely Easy**
			Frequency (%)		
Difficulty of Building Trust	1 (4)	10 (36)	10 (36)	6 (21)	1 (5)
Difficulty of Building Relationships	2 (7)	15 (54)	4 (14)	6 (21)	1 (4)

Downloaded from https://academic.oup.com/iwc/advance-article/doi/10.1093/iwc/iwad029/7136028 by Karlstads universitet user on 12 May 2023

Table 3. Importance of Trust and the amount of time spent in global virtual team (GVT) activities

		% of Time spent in Virtual Team Activities		
		Less than One-Third	Between One & Two-Thirds	More than Two-Thirds
		Frequency (%)		
Importance of Trust	Slightly important	1 (4)	0 (–)	0 (–)
	Very Important	3 (11)	4 (15)	6 (22)
	Extremely Important	0 (–)	7 (26)	6 (22)
Total		4 (15)	11 (41)	12 (44)

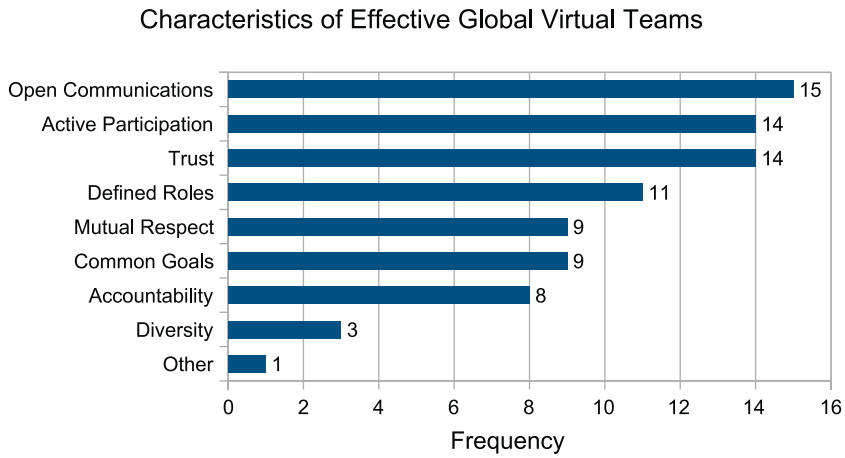

Figure 1. Characteristics of Effective Global Virtual Teams (GVTs).

Table 4. Challenges for Global Virtual Teams (GVTs)

Challenges for Global Virtual Teams	Frequency (%)
Time Zones	19 (68)
Lack of personal Connection	19 (68)
Communication Failures	17 (61)
Work Ethics	7 (25)
Technology	4 (14)
Culture Differences	3 (11)

between the difficulty of building relationships and the difficulty of knowledge sharing (ρ, $p \leq 0.01$). The association between the difficulty of building a cohesive team and the difficulty of knowledge gathering is significant (ρ, $p \leq 0.01$). Those older than 25 years report that it is more difficult to build a cohesive team than those under 25 (Chi Sq, $p \leq 0.05$). Gender or the amount of time spent in GVT activities does not influence this result.

Trust and Relationships: Trust is important for members of GVTs. Its importance correlates (ρ, $p \leq 0.01$, 1-tailed) with the importance of building relationships and the importance of informal conversations as well as creating a healthy culture and being creative (ρ, $p \leq 0.05$, 1-tailed). This result is interpreted to say that trust is needed to build relationships, and informal conversations help in building trust. Trust also helps in creating a healthy culture and for the GVT to be creative.

As anticipated, the level of trust and the strength of relationships seem to impact on the ability to share or gather knowledge. There is a significant association between building trust and building relationships (ρ, $p \leq 0.01$, 1-tailed). In other words, the more difficult it is to build trust, the more difficult it is to build relationships. There are also significant associations between trust and knowledge sharing and between trust and knowledge gathering (ρ, $p \leq 0.01$, 1-tailed) which is interpreted to mean that if the level of trust is weak, then it is more difficult to gather or share knowledge. Difficulties in building trust is also linked to difficulties in resolving conflict (ρ, $p \leq 0.01$, 1-tailed) and difficulties in reaching agreements (ρ, $p \leq 0.05$, 1-tailed). As one might predict, the greater the difficulties experienced in building relationships, the greater the difficulty experienced in reaching agreements and building a healthy culture (ρ, $p \leq 0.01$, 1-tailed). Resolving conflict is also difficult when there are difficulties in building the relationship in a virtual environment (ρ, $p \leq 0.05$, 1-tailed). Differences in gender, age group or the time spent in GVT activities do not influence these views associated with trust and relationships.

3.1.3. *Conflict resolution and agreements*

Conflict inhibits effective teamwork. As one might expect, conflict creates difficulties in knowledge gathering, building trust and reaching agreements (ρ, $p \leq 0.01$, 1-tailed) as well as making relationship building more difficult (ρ, $p \leq 0.05$, 1-tailed). The fact that resolving conflict is one of the most difficult activities for GVTs emphasises the importance of conflict resolution for GVTs so that they can be effective. As discussed in section 3.2, it can be difficult for managers to recognise non-overt conflict and to know when, or if, an overt conflict episode has been resolved.

3.1.4. *Defined roles and accountability*

Our results place less emphasis than expected on defined roles and accountability in GVTs, which is surprising given that, according to Katzenbach and Smith's (1993, p.45) definition '… a team is a small number of people with complementary skills who are committed to a common purpose, performance goals and approach for which they hold themselves mutually accountable.'

Downloaded from https://academic.oup.com/iwc/advance-article/doi/10.1093/iwc/iwad029/7136028 by Karlstads universitet user on 12 May 2023

Table 5. Difficult Activities for Global Virtual Teams (GVTs)

	Extremely Difficult	Somewhat Difficult	Neither Easy nor Difficult	Somewhat Easy	Extremely Easy
			Frequency (%)		
Building a Cohesive Team	1 (4)	13 (48)	5 (19)	8 (30)	0 (0)
Resolving Conflict	1 (4)	12 (43)	8 (29)	7 (25)	0 (0)
Creating a Healthy Culture	2 (7)	12 (43)	7 (25)	7 (25)	0 (0)
Reaching Agreements	0 (0)	6 (21)	12 (43)	9 (32)	1 (4)
Being Creative	2 (7)	9 (32)	8 (29)	7 (25)	2 (7)

In healthcare, for example, focusing on accountability in teamwork is demonstrated to improve patient outcomes (Kelly and Faraone, 2013). In our results, accountability is not amongst the top three characteristics of GVTs with only 29% of participants raising accountability as an attribute.

3.1.5. *Gender and age*

Gender: Females are finding fewer communication issues in a remote environment. Nine of the 16 male respondents (56%) say that remote communications are more difficult; however only three of the 11 female respondents (27%) report more difficulty. While the result suggests that a possible difference between genders, it is not statistically significant. The only difference that gender makes is in relation to the use of voice calls (and reported in section 3.1.1): females report using more voice calls than males, even though voice calls are not necessarily their preference.

Females report knowledge sharing to be easier than males with 6/11 female respondents (55%) saying that knowledge sharing is easy in a remote environment compared to 4/16 male respondents (25%), While this result is not statistically significant, it prompts further study as the numbers are low.

Age: there are differences noted when the data are analysed based on respondents Age. Younger respondents seem to adapt easier to virtual work and are more in favour of collaborative activities. Although the sample size is small with only three respondents (11%) under the age of 25, all three say that virtual projects can be as effective as collocated projects and that virtual work can be as rewarding as collocated work. All three respondents in this group choose active participation and open communications in the top three characteristics for building an effective team. The fact that those older than 25 years who spend more than two-thirds of their time in GVT activity report more difficulty than those under 25 years (Chi Sq $p \leq 0.05$) may be a reflection of their greater work experience and involvement in more projects.

For the greater than 25 year old age group, less than half (10, 40%) believe that virtual work can be as effective as collocated work and 12 (48%) of these older than 25 year olds agree that virtual work can be as rewarding as collocated work. For building an effective team, the joint top characteristics are trust (12, 48%) and open communications (12, 48%) and the third is active participation (11, 44%).

Age does not influence other beliefs or behaviours; there is no statistically significant difference between Age category and the other variables. More data are needed to check if there are differences in attitudes amongst different age groups.

3.2. Findings from interviews

Technology is identified by four of the five interviewees when asked for the characteristics of GVTs. All five of the interviewees acknowledge the importance of technology and the suitability of the current software available for day-to-day virtual teamwork. Several new communication tools and technology have been introduced in recent years.

The technology in use now 'is an improvement'[INT1] and 'it is much better'[INT1] but it is also pointed out that technology mediated communication is not the ideal way to communicate and INT1 says 'we have to adapt, it's assumed the way to go forward' [INT1]. Robinson (2021) suggests the technology we have today may not be suitable for GVTs and that we are learning to change as we go in order to be productive.

There are conflicting responses to using the existing technology for knowledge sharing and knowledge gathering. People tend to assume they have provided all the required information when sharing with their team members through whichever media is being used, e.g. email or even video calls. However, the receiver who is trying to gather knowledge may not always understand what is being shared. INT4 says that training is more difficult virtually and 'takes longer than if you are in the office' [INT4]. INT3 suggests that people have different learning styles with some preferring traditional methods such as watching video demos or using shared documentation but those methods may not work for those who prefer more interactive or 'hands-on' approach, which is more difficult in a virtual environment.

The appropriate use of existing technology seems to be situation dependent. Email remains the most widely used form for communication with people preferring email for its quick and convenient nature. However, there is potential to deem a task complete once an email has been sent and a certain level of trust is required between the sender and receiver that the contents of the email have been understood or if not then questioned by return with INT1 saying ' you have no idea' if the message is being acted upon and 'having to follow up with the person' to get clarity which can lead to delays. INT4 prefers email as it is effectively a record or 'paper trail' of the communication which could suggest a lack of trust with the receiver. INT2 added that email is a handy method to send a quick message almost as a reminder 'before I forget' but with the expectation that it could be a couple of days before getting a response so is not ideal for important communication unless 'proof' of the conversation happening is required. Portraying the correct level of urgency in a non-face-to-face communication is difficult, with email or texting being used to share the problem or potentially pass the problem on to someone else knowing that a follow up conversation will be required to clarify the message being communicated.

Video calls can also pose challenges when used in situations where active participation is required. Company-wide or larger meetings where participants may be passive can be effective using existing technologies, but such communication methods are not as appropriate when active participation is expected. Limiting the number of attendees to give equal opportunity to all to participate can make better use of the meeting with INT3 suggesting that video calls can sometimes be dominated by some people while other people stay quiet in the background, just listening' [INT3].

Downloaded from https://academic.oup.com/iwc/advance-article/doi/10.1093/iwc/iwad029/7136028 by Karlstads universitet user on 12 May 2023

Although without visual clues it is impossible to know who is engaged in the call. INT1 shares the view that on video calls sometimes 'people are slow to put up their hand and can get left behind' [INT1], while also suggesting that some calls get dominated by one or two people who ask all the questions while others stay quiet. INT5 says that people can become disengaged and lose focus if there are a lot of people on a call thus reducing the value of the call.

The use of any virtual communication is not suitable in situations for training [INT4] or for brainstorming or project kick-offs [INT3]. Although the survey results indicate that people can contact their teammates when required it is not evident if the methods being used are efficient for the task.

The lack of visual clues is noted as a big challenge in communicating today. Excluding video calls, all other forms of communication being used cannot replicate the face-to-face communication that provide the non-verbal aspects of a conversation. INT5 especially notes facial expressions as being important and being able to see the person to whom you are talking can help with relationship building which is something missing from email or other modes of communication. INT1 also says that 'you can get an idea of what the other person is thinking through their body language and facial expressions' [INT1], when face-to-face or on video calls.

We must acknowledge that individuals have varying communication styles, with some being more visual and who will always use video calls with cameras turned on, while others may prefer the quick and easy email approach. People seem reluctant to enable cameras on long meetings or calls with many participants which could suggest they are not fully engaged in the meeting and may be conducting other activities at the same time. INT5 observes that 'you definitely pay attention' when cameras are turned on.

Informal or 'water cooler' conversations are deemed to very important. At the beginning of the Covid-19 pandemic, team members engaged with each other through occasional team quiz nights over Zoom or another application. Although primarily set up for the social aspect, these meetings also attempted to replicate the water cooler conversations with occasional pieces of important work-related information being exchanged. As the pandemic continued these conversations petered out with everyone focused on work and other activities. INT1 noted that 'it was a novelty at the start but the novelty wore off'.

Informal conversations tend to be important for work related information sharing. INT4 reported that informal chats are a 'great place to get a lot of info and perspective' and to 'pick up a lot of information' [INT4]. As described by INT1, water cooler conversations are very important for relationship building and are a great way to 'build up rapport' with teammates and colleagues, that can ultimately lead to things getting done quicker, due to having the relationship already in place. INT5 raised the point that without water cooler conversations we are not having the 'same relationships with people'. Such informal conversations can increase cohesion within the team (Bond-Barnard *et al.*, 2018).

The topic of conflict brings mixed views with the initial consensus being that conflict can be resolved within GVTs in the same way as it can be resolved in a face-to-face team. Although questions remain on how to recognise that conflict exists in the first place and on how to know if it has been resolved. INT3 describes the difficulty of recognizing that conflict exists because 'you can't see the other person' [INT3] and we are missing the visual clues. Different writing or communication styles, and in particular the tone of communications, may also be misinterpreted giving a false impression that conflict may (or may not) exist.

Resolving conflict in a virtual team can typically be addressed by normal channels such as manager intervention or over phone calls which INT3 says 'may be easier for some people' if they are virtual and could find a face-to-face conversation difficult.

It is difficult to know if conflict has been resolved in a virtual environment. INT4 reports that 'it [conflict] can continue to bubble under the surface' when we are all virtual or remote. Our findings align with research by Olson and Olson (2000) who suggest that it is easy for strained relationships to form in a virtual setting and Scott-Young and Samson's (2008) finding that there are delays in conflict resolution the further teams are apart.

Building trust and relationships go hand-in-hand; they have a significant impact on the cohesion of a team and how the team performs. From the survey analysis we can see the importance of trust and the difficulty in building trust. There is a strong association identified between trust and relationship building, see Table 2. Team members are working in isolation with much less interaction between one another, particularly a lot fewer casual interactions, making relationship building much more difficult. The lack of visual interactions is an impediment to team cohesion with INT5 reporting that 'without having video calls, it's very hard to make a relationship' [INT5]. INT1 describes that having a 'visual rapport' when you can see the other person face-to-face on a video call 'works much, much better' [INT1].

Having cameras turned on helps with building trust and relationships through the visual nature of the communication. INT4 notes that enabling their cameras has been made mandatory for all of the team at their weekly team meetings. The visual communication via camera helps with team bonding and team building; and having a quick catch-up before going straight into the meat of the conversation is seen to be important to maintain and develop the relationship between the team members. Making a small change to the daily routine, such as enabling cameras for team meetings, can add to the relationship between work processes and trust (Powell *et al.*, 2006).

The social aspect may be the most important part of being collocated along with the informal conversations it enables, and some work activities are best performed in a face-to-face environment. INT1 remarked that it can be more productive when face-to-face and '... can get things done quicker'. This feeling is shared by INT3 who also believes that certain tasks can be performed quicker and in a more productive manner when in a face-to-face environment. INT4 highlights the advantages of having a flexible solution but also stresses that there are 'certain times that everybody should be together' [INT4], depending on the task or activity in question.

Our findings indicate that people are adapting to work in GVTs and believe they can be as effective as collocated teams using the technology that is currently available. Even though people agree that email is not the best or their preferred method for virtual communications, it is, nonetheless, the most widely used means given the quick and convenient nature of email. However, communications failures are noted as being one of the top three challenges affecting GVTs with all respondents agreeing that it is important to know that your message is received and understood. The lack of personal connection and having much fewer informal conversations is impacting on areas of team cohesion such as building trust and resolving conflict, with knowledge gathering and knowledge sharing also proving difficult in GVTs.

Downloaded from https://academic.oup.com/iwc/advance-article/doi/10.1093/iwc/iwad029/7136028 by Karlstads universitet user on 12 May 2023

4. DISCUSSION

We aim to identify the main challenges that GVTs encounter particularly during the Covid-19 pandemic when working from home became mandatory. Common themes such as time zones and cultural differences are identified as challenges affecting GVTs (Morrison-Smith and Ruiz, 2020a). We also identify the lack of personal connection as a new and significant challenge given the enforced remote nature of GVTs because of the Covid-19 pandemic. As organizations move forward with plans for the future of work, social and personal contact will continue to be a challenge for members of GVTs.

Communications failure is an additional challenge that we identify, as GVTs become more reliant on technology to communicate. Asynchronous communication methods such as email or text messaging are widely used by virtual team members, primarily given their quick and convenient nature. However, the use of asynchronous forms of media to communicate eliminates the non-verbal clues and makes it difficult to get feedback to know if messages are being received and understood. The lack of immediate responses with asynchronous communication methods can lead to delays and confusion in understanding the message. Often additional communications are required to clarify the messages being sent. Previous research suggests that without a response to a message, the sender can only assume it has been understood (Suchan and Hayzak, 2001). Communications failure is raised in this study as being particularly prevalent in training situations or with new hire induction where previously classroom or one-to-one sessions would have been conducted giving immediate possibilities to provide feedback and to ask for clarity.

Bond-Barnard *et al.* (2018) suggest that collocated teams have the advantage of increased frequency of communication that physical proximity affords, which helps to promote team cohesion and collaboration. Similarly we find that the lack of informal communications such as the traditional 'water cooler' conversations, is a challenge that impacts GVTs. Informal conversations are noted by all interviewees as being very important to help build and maintain communication channels. As well as being used for genuinely informal reasons, informal collated encounters are ideal for passing on little nuggets of information, and greetings, that people may not otherwise exchange and are a significant aspect of relationship building.

We also investigate if teams can build cohesion when interactions are virtual. Building a cohesive team is vital for the team to function to its potential. Trust and relationships are significant aspects of any cohesive team; indeed, individual trust and team cohesion share a reciprocal impact on each other that improve overall performance (Paul *et al.*, 2016). In other words the more individual team members trust one another, the greater the team cohesion, which further promotes more individual trust.

However, trust cannot exist without firstly building a relationship and, although there are complexities, the longer the relationship exists the greater the level of trust that can be built (Vanneste *et al.*, 2014). The use of the appropriate communications technology has a direct impact on team members' ability to build trust and relationships. Indeed, Gibson and Cohen (2003) suggest that delays or slow responses to other team members communications can lead to mistrust, although feedback from our interviews suggest that people sometimes expect delays, primarily due to the nature of email.

From the survey data, there is a significant link between 'building trust' and the 'difficulty in reaching agreements' ($p \leq 0.01$). A lack of trust makes it more difficult to reach agreement. Data from the interviews further highlight that it can sometimes take longer to complete tasks virtually, which is in part due to the lack of a relationship with other team members.

Our research identifies that knowledge sharing in a virtual environment is easier than knowledge gathering. Sending an email for example, is sometimes considered as knowledge sharing but the ability to read and understand the details (i.e. knowledge gathering) is challenging. Different writing styles can impact the sender's ability to convey the message in a format that the receiver can comprehend. Greater difficulty in gathering or understanding information is found for other modes of communication too, even video calls, where the sender and receiver can be on different personal 'wavelengths' due to their communication styles and ability to deliver or receive messages. Inefficient communications can impact on the team's ability to function effectively (Morgan *et al.*, 2014), and distance between the parties in a conversation is one aspect to consider when choosing the appropriate communications media (Sivunen and Valo, 2006).

While conflict is often thought of as being a bad thing, it can have a positive effect and is a necessary attribute to create a healthy team. The storming aspect of team development where conflict is typically addressed is not so important in virtual teams (Johnson *et al.*, 2002). However, a recurring theme in our results is the fact that it can be difficult to actually recognise when there is conflict in a virtual team. Conflict can go unnoticed and bubble under the surface between one or more team members. If not identified then it cannot be addressed and can impact on the team's performance. Our results suggest that resolving conflict may be straightforward through a manager's intervention using telephone or video calls but that it will always be difficult to know if the conflict has been effectively resolved when there are no visual clues.

Defined roles and teamwork go hand-in-hand since teams typically include a mix of specialized knowledge and skills, and team members hold themselves mutually accountable (Katzenbach and Smith, 1993). It is therefore somewhat surprising that in our study neither the issue of defined roles, nor that of accountability engage more than 29% of our participants. GVTs face additional challenges and dilemmas that are not experienced in collocated settings. Although part of a team, individuals in a GVT are isolated and notions of accountability can vary according to whether the culture is more collective or individualistic (Dube and Paré, 2001). Complex interdependent behaviour patterns exist when a task is completed through collaborative teamwork (Fandt, 1991). Because individual GVT members, although isolated, do not work in isolation, the joint effects of accountability and interdependent behaviours could be considered as potential enhancements to the team effort (Ebrahim *et al.*, 2009). Indeed, virtual team members must have clear roles and accountabilities and a lack of visibility may cause virtual team members to feel less accountable for their results (Ebrahim *et al.*, 2009). Accountability and responsibility are considered amongst the main contributing factors to the commitment to team success and employee relations (Horwitz *et al.*, 2006). Further study is indicated to investigate if GVTs generally exhibit less accountability than other teams and to investigate if particular factors may influence the level of accountability within GVTs.

Teamwork is an essential component of contemporary organizations, and effective teams are promoted as a means of enhancing organizational performance (Mickan and Rodger, 2000). Our results show that communication in GVTs is critically important

Downloaded from https://academic.oup.com/iwc/advance-article/doi/10.1093/iwc/iwad029/7136028 by Karlstads universitet user on 12 May 2023

and key to building and maintaining high levels of cohesion. Similar to our results, Robinson (2020) shows that the technology currently available is generally sufficient in supporting virtual team communications needs for work related activities. However, we identify that the 'softer' communications requirements such as the informal and social needs of team members are not adequately supported by the technology currently available. Our interviewees report that team members are participating in much fewer informal communications and that this lack of informal interaction is adversely affecting relationships within the teams.

5. CONCLUSIONS

Our research, as well as validating findings in previously published literature, contributes additional insights into the challenges facing members of GVTs, particularly during the Covid-19 pandemic. This research also provides insights into issues surrounding virtual team cohesion and suggests a number of key recommendations which should prove beneficial to the deployment and the development of GVTs.

There is a need to adapt to virtual work and the results show that since the early months of 2020, people have been adapting and changing the way they work by learning as they go. The existing software tools we have available and ways of communicating need to change (Robinson, 2020). Work needs to become more collaborative and interactive in nature to ensure relationships can be maintained, which will positively impact the quality of work and happiness of workers. Twenty years after Olson & Olson's work, it is still true that change in work practices is required for GVTs to be able to collaborate successfully (Olson and Olson, 2000).

GVTs have been in existence for some time and their use is expected to increase in the coming years. Attempts need to be made to try and make virtual work encompass the positive traits of collocated work as much as possible, with our results putting emphasis on the social side of work. Being collocated allows for increased levels of communication between team members, both formal and informal, and thus promotes greater amounts of collaboration and interaction. Achieving the same levels of collaboration and interaction virtually may be an impossible goal, but it is none the less a goal we should strive to achieve.

A possible limitation of this research is that it is conducted within a single organization. Although the organization in question can be considered to be representative of the chosen industry, other organizations may have implemented different work processes or use different tools and technology to assist their work force deal with working in GVTs. An industry wide research project could prove beneficial to further enhance the validity of the findings. However, the sample size represents almost 50% of the target population, so our results are important.

The Covid-19 pandemic itself added to the limitations of this research, in particular to the interviews which were all conducted over Zoom calls. In person interviews may have provided additional information through non-verbal clues or the body language of the participants which might have directed conversations differently and added to the data collected.

5.1. Recommendations

Considering our survey and interview results in the context of related research there are some immediate recommendations that are evident for the successful operation of GVTs.

Communications: The use of synchronous forms of communications should be promoted whenever possible to reduce 'communications failure' which is identified in the results as one of the top three challenges in GVTs. Synchronous communication tends to be faster, leaves less room for delays and confusion and creates a better working environment. Face-to-face is considered the richest medium for communication (Windsperger and Gorovaia, 2010), but when teams are virtual the next appropriate media are video calls followed by voice calls. In group settings, new or additional types of virtual meeting training could help in executing more efficient online meetings to ensure active participation and improved levels of engagement, both of which would help with knowledge gathering and sharing which are identified as particular challenges to building team cohesion.

The use of asynchronous communication such as email, which the results show is the most used form of communication, should be limited in use to less important communications or for communications that need to be written, such as when a sign-off is required on a task or deadline. It is important to know that your message has been received and understood so the appropriate media should be used, depending on the situation. In training situations or when knowledge sharing or gathering is the task in hand, media such as video calls should be used so that visual and non-verbal clues can be available to help reduce communications failures. Richer forms of media also offer synchronous communication to GVTs which reduce the possibility of failures.

Our results show that the social aspect should not be forgotten. There is a tendency to put greater importance on our contacts and their messages when the social contact is face to face (Elron and Vigoda-Gadot, 2006). We show that informal connections can contribute to both personal and business relationships and are an important attribute for future communications. Achieving the same level of social contact in a virtual team as in a face-to-face team will prove very difficult but efforts should be made to promote informal activities to help maintain as high a level of social contact as possible.

Team Development and Building Cohesion: A number of factors impact on building a cohesive team that are true in both face-to-face and in GVTs. However, the remote nature of GVTs makes building trust and relationships more difficult. The use of interactive forms of communications media and promoting more collaborative and cooperative work could help to develop high performing GVTs. Leaving the Covid-19 pandemic aside, the use of face-to-face for initial interactions such as introductory meetings or project kick-offs should be explored given the importance of such interactions for laying the foundations to building relationships. Interactive sessions are more appropriate where brainstorming is required, which achieves better engagement in a face-to-face environment. Interactive and/or engaging activities would benefit from the use of more collaborative tools and processes when in a virtual environment.

Knowledge is a strategic resource to any organization, and we show that finding better ways to create, share and gather knowledge can prove beneficial to GVTs. In collocated teams, knowledge is created naturally through the shared understanding of the team members (Gibson and Cohen, 2003). However, shared understanding is not as evident in GVTs compared to collocated teams and ways need to be found to recreate the understanding of team characteristics such as goals and tasks.

These results highlight that the use of traditional methods for knowledge sharing and gathering are inefficient in GVTs and immersive technologies should be investigated to provide mechanisms to make the process more interactive. By enabling a

Downloaded from https://academic.oup.com/iwc/advance-article/doi/10.1093/iwc/iwad029/7136028 by Karlstads universitet user on 12 May 2023

two-way communications channel, both the sharer and gatherer of the knowledge can benefit from real time feedback and communications.

Recognizing and resolving conflict can also prove to be more difficult in GVTs according to our informants. Using synchronous forms of communication can help limit conflict or at least limit the potential for conflict to arise. Differing communication styles can impact peoples' perceptions of others.

Body language and visual clues can quite often help eliminate any poor perceptions. Implementing more collaborate work practices or creating mini-teams within teams for work sharing and shadowing also can promote better team bonding and reduce conflict and enable more social conversations.

6. FUTURE WORK

This research was conducted just over 1 year after the Covid-19 pandemic had begun. At that point there was little newly published research into the communications challenges affecting GVTs, in particular when the choice to be remote was not voluntary. As the world evolves and more focus is put on the future of work in regard to where we work and organizations have the time to implement new ways of working (and the tools and processes to support it) future, additional research could yield different findings. As well as organizations changing how they work, individuals will also change. We show that individuals are adapting to working in GVTs, so further research will be beneficial to see the additional challenges that working in GVTs will bring and if individuals are still evolving in a positive manner.

Results on gender differences in the use of different modes of communication merits further investigation to understand why there is a difference between the preferred mode of communication, and their use, for different genders. The reason why females use more voice calls remains unexplained.

Additional research comparing the experiences of different age groupings could yield new information. 10% of the participant (three) are under the age of 25, where working in a virtual team may have been their first and only employment experience. Although we show little difference in the challenges affecting different age groups, more research and analysis of the younger demographic could produce useful results for future discussion.

Given the constraints brought about by the Covid-19 pandemic, GVTs (like others) had to accept working in less than ideal circumstances. The experience of coping with the existing technology and appropriating it to serve the immediate needs of the GVT highlighted limitations that should be addressed in the design of future technologies. As well as recognizing that global virtual teamwork is expanding, we must also acknowledge that further pandemics or international crises are predicted that will force us to immediately redesign our work practices. We can learn from our recent experience and be better prepared for the unseen and unpredictable circumstances of the future. By designing better technologies we can predict that we will improve the work experience and productivity of GVTs in the process.

References

Bond-Barnard, T. J., Fletcher, L. and Steyn, H. (2018) Linking trust and collaboration in project teams to project management success. *Int. J. Manag. Proj. Bus.*, **11**, 432–257.

Dakory, M. and Abdou, H. (2009) Virtual teams processes: a conceptualization and application. *Probl. Perspect. Manag.*, **7**, 15–26.

Dube, L. and Paré, G. (2001) Global virtual teams. *Commun. ACM*, **44**, 71–73.

Ebrahim, N. A., Ahmed, S. and Taha, Z. (2009) Virtual teams: a literature review. *Aust. J. Basic Appl. Sci.*, **3**, 2653–2669.

Elron, E. and Vigoda-Gadot, E. (2006) Influence and political processes in cyberspace: the case of global virtual teams. *Int. J. Cross-cult. Manag.*, **6**, 295–317.

Fandt, P. M. (1991) The relationship of accountability and interdependent behavior to enhancing team consequences. *Group Organ. Stud.*, **16**, 300–312.

Gibson, C. B. and Cohen, S. G. (2003) *Virtual teams that work: Creating conditions for virtual team effectiveness*. John Wiley & Sons, San Francisco.

Horwitz, F. M., Bravington, D. and Silvis, U. (2006) The promise of virtual teams: identifying key factors in effectiveness and failure. *J. Eur. Ind. Train.*, **30**, 472–494.

Johnson, S. D., Suriya, C., Yoon, S. W., Berrett, J. V. and La Fleur (2002) Team development and group processes of virtual learning teams. *Comput. Educ.*, **39**, 379–393.

Katzenbach, J. R. and Smith, D. K. (1993) The rules for managing cross-functional reengineering teams. *Plan. Rev.*, **21**, 12–13.

Kelly, S. and Faraone, L. (2013) Improving the ed experience with service excellence focused on teamwork and accountability. *J. Emerg. Nurs.*, **39**, 33–36.

Mickan, S. and Rodger, S. (2000) The organisational context for teamwork: comparing health care and business literature. *Aust. Health Rev.*, **23**, 179–192.

Morgan, L., Paucar-Caceres, A. and Wright, G. (2014) Leading effective global virtual teams: the consequences of methods of communication. *Syst. Pract. Action Res.*, **27**, 607–624.

Morrison-Smith, S. and Ruiz, J. (2020a ISSN 2523-3971) Challenges and barriers in virtual teams: a literature review. *SN Applied Sciences*, **2**, 1096. https://doi.org/10.1007/s42452-020-2801-5.

Morrison-Smith, S. and Ruiz, J. (2020b) Challenges and barriers in virtual teams: a literature review. *SN. Appl. Sci.*, **2**, 1–33.

Olson, G. M. and Olson, J. S. (2000) Distance matters, chapter Face-to-Face Group Work compared to Remote Group Work with and without Video. In Sellen, A. J., Finn, K. E., Wilbur, S. B. (eds), *Video-Mediated Communication*. ACM Press.

Paul, R., Drake, J. R. and Liang, H. (2016) Global virtual team performance: the effect of coordination effectiveness, trust, and team cohesion. *IEEE Trans. Prof. Commun.*, **59**, 186–202.

Peters, L. M. and Manz, C. C. (2007) Identifying antecedents of virtual team collaboration. *Team performance management: an. Int. J.*, **13**, 117–129. https://doi.org/https://www.emerald.com/insight/content/doi/10.1108/13527590710759865/full/html.

Powell, A., Galvin, J. and Piccoli, G. (2006) Antecedents to team member commitment from near and far: a comparison between collocated and virtual teams. *Inf. Technol. People*, **19**, **19**, 299–322. https://doi.org/10.1108/09593840610718018.

Robinson, B. (2020) *How remote working is reshaping a future new world of work*. Forbes.

Bryan Robinson. How Remote Working Is Reshaping A Future New World Of Work. In *Working Remote*. Forbes, 2021. URL https://www.forbes.com/sites/bryanrobinson/2020/05/04/how-remote-working-is-reshaping-a-future-new-world-of-work/ .

Scott, G. G. and Fullwood, C. (2020) Does recent research evidence support the hyperpersonal model of online impression management? *Curr. Opin. Psychol.*, **36**, 106–111.

Scott-Young, C. and Samson, D. (2008) Project success and project team management. *J. Oper. Manag.*, **26**, 749–766.

Sivunen, A. and Valo, M. (2006) Team leaders' technology choice in virtual teams. *IEEE Trans. Prof. Commun.*, **49**, 57–68.

Downloaded from https://academic.oup.com/iwc/advance-article/doi/10.1093/iwc/iwad029/7136028 by Karlstads universitet user on 12 May 2023

Suchan, J. and Hayzak, G. (2001) The communication characteristics of virtual teams: a case study. *IEEE Trans. Prof. Commun.*, **44**, 174–186.

Tuckman, B. W. and Jensen, M. A. C. (1977) Stages of small-group development revisited. *Group Organ. Stud.*, **2**, 419–427.

Vanneste, B. S., Puranam, P. and Kretschmer, T. (2014) Trust over time in exchange relationships: meta-analysis and theory. *Strateg. Manag. J.*, **35**, 1891–1902.

Windsperger, J. and Gorovaia, N. (2010) Knowledge attributes and the choice of knowledge transfer mechanism in networks: the case of franchising. *J. Manag. Gov.*, **15**, 617–640.

Downloaded from https://academic.oup.com/iwc/advance-article/doi/10.1093/iwc/iwad029/7136028 by Karlstads universitet user on 12 May 2023